

CSE 331

AUTOMATA AND COMPUTABILITY

Lecture 3

Spring 2026

Today's learning goals

Sipser Ch 1.1, 1.2

- Review what it means for a **set** to be closed under an **operation**.
- Define the regular operations on languages
- Prove closure properties of the class of regular languages

Building DFA



Remember

States are our only (computer) memory.

Design and pick states with specific roles / tasks in mind.

"Have not seen any of desired pattern yet"

"Trap state"

A useful (optional) bit of terminology

When is a string accepted by a DFA?

$$M = \{w : \delta^*(q_0, w) \in F\}$$

$$\delta(q, c) = q_{\text{next}} \\ q \in Q \quad c \in \Sigma$$

Computation of M on w : *where do we land when start at q_0 and read each symbol of w one-at-a time?*

$$\delta^*(q, w) = \begin{cases} q & \text{if } w = \varepsilon \\ \delta(q, c) & \text{if } w = c, c \in \Sigma \\ \delta^*(\delta(q, c), w') & \text{if } w = cw', c \in \Sigma, w' \in \Sigma^* \end{cases}$$

Recursively
defined
function

Regular languages

Sipser p. 35 Def 1.5

- For DFA M over the alphabet Σ
 - For each string w over Σ , M either accepts w or rejects w
 - The **language recognized by M** is the set of strings M accepts
 - The **language of M** is the set of strings M accepts
 - **$L(M)$** = $\{ w \mid w \text{ is a string over } \Sigma \text{ and } M \text{ accepts } w \}$

Definition: A language is **regular** iff there is some finite automaton that recognizes **exactly** it.

Regular languages: bounds?

Is **every** finite language regular?

finitely Many Strings

- A. No: some finite languages are regular, and some are not.
- B. No: there are no finite regular languages.
- ☒ C. Yes: every finite language is regular.
- D. I don't know.

*DFA has one path for each String, and
Since there are Only finitely many Strings,
DFA has finitely many States.*

Building DFA



New strategy

Express L in terms of simpler languages – use them as building blocks.

Example

$$\begin{aligned} L &= \{ w \mid w \text{ does not contain the substring baba} \} \\ &= \text{the complement of the set} \\ &\quad \{w \mid w \text{ contains the substring baba} \} \end{aligned}$$

Substring, prefix, suffix

If and only if

V is a Substring of $w \iff$ there exist strings x, y such that
 $w = xvy$.

if $x = \epsilon$, then v is prefix of w .

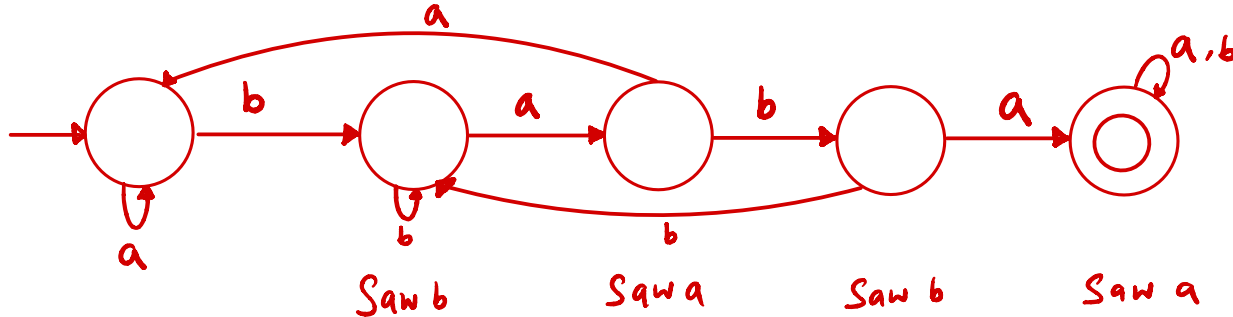
if $y = \epsilon$, then v is suffix of w .

Building DFA



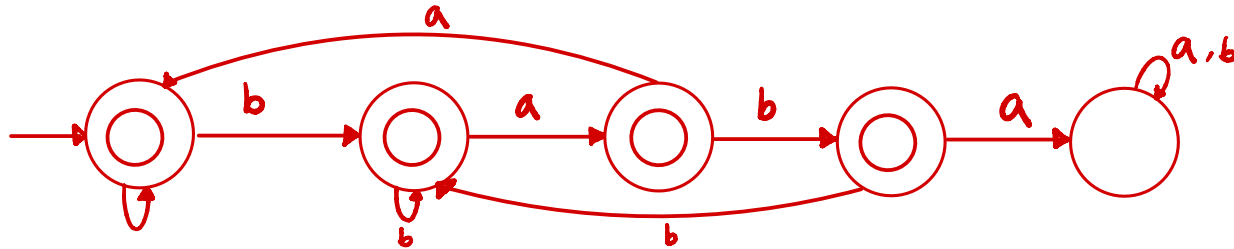
DFA recognizing

$L = \{w \mid w \text{ contains the substring baba}\}$



DFA recognizing

$L^c = \{w \mid w \text{ doesn't contain the substring baba}\}$



Complementation

Theorem: If A is a regular language over $\{0,1\}^*$, then so is its complement $\overline{A}$

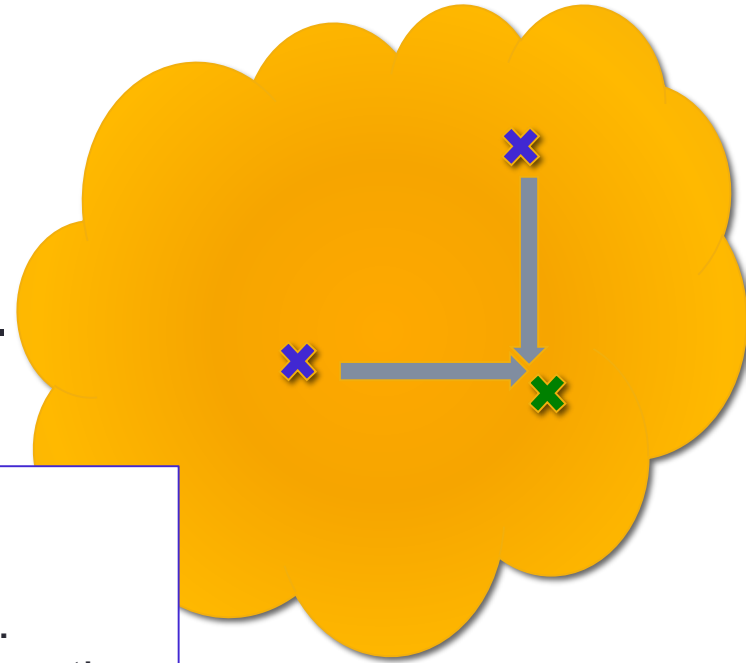
aka "the class of regular languages is closed under complementation"

Closure of ... under ...

- $\mathbb{Z}$ under addition.
- Set of even ints under multiplication.
- $\{0\}^*$ under concatenation.

Which of these is true?

- A. The set of odd integers is closed under addition.
- B. The set of positive integers is closed under subtraction.
- ☒ C. The set of rational numbers is closed under multiplication.
- D. The set of real numbers is closed under division.
- E. I don't know.



Rational number can
be written as $\frac{a}{b}$
Where $a, b \in \mathbb{Z}$ (integer)

Complementation

Theorem: If A is a regular language over $\{0,1\}^*$, then so is its complement $\overline{A}$

aka "the class of regular languages is closed under complementation"

Complementation

$\bar{A}, A^c$
Complement

Theorem: If A is a regular language over $\{0,1\}^*$, then so is its complement $\bar{A}$

aka "the class of regular languages is closed under complementation"

Proof: Let A be a regular language.

Then there is a DFA $M=(Q,\Sigma,\delta,q_0,F)$ such that $L(M) = A$.

We want to build a DFA M' whose language is $\bar{A}$. Define

$M' =$

?

Claim of Correctness $L(M') = \bar{A}$

Proof of claim...

A is regular, accepted by DFA $M = (Q, \Sigma, \delta, q_0, F)$

Define new DFA $M' = (Q, \Sigma, \delta, q_0, Q \setminus F)$ ← Construction

↳ Set of states without F,
this is known as
difference between sets.

Claim: M' accepts A^c

Proof: because M accepts A

$$A = \{w : M \text{ accepts } w\} = \{w : \delta^*(q_0, w) \in F\}$$

$$A^c = \{w : w \notin A\} = \{w : \delta^*(q_0, w) \notin F\} = \{w : \delta^*(q_0, w) \in Q \setminus F\}$$

So M' accepts A^c ■

Claim: M' accepts A^c

Proof: because M accepts A

$$A = \{w : M \text{ accepts } w\} = \{w : \delta^*(q_0, w) \in \underline{F}\}$$

$$A^c = \{w : w \notin A\} = \{w : \delta^*(q_0, w) \notin F\} = \{w : \delta^*(q_0, w) \in \underline{Q/F}\}$$

So M' accepts A^c ■

Correctness

Why closure proofs?

- General technique of proving a new language is regular
- Stretch the power of the model
- Puzzle!

The regular operations

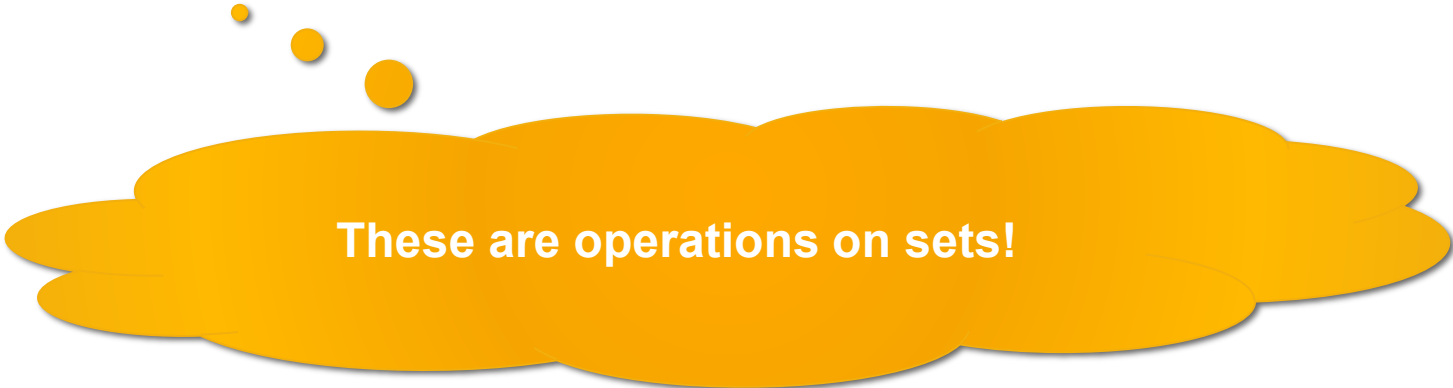
Sipser Def 1.23 p. 44

For A, B languages over same alphabet, define:

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\} \quad \text{Today}$$

$$A \circ B = \{xy \mid x \in A \text{ and } y \in B\}$$

$$A^* = \{x_1 x_2 \dots x_k \mid k \geq 0 \text{ and each } x_i \in A\}$$



These are operations on sets!

Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

If A, B are regular languages, then $A \cup B$ is also regular.

Proof:

What are we proving here?

- A. For any set A , if A is regular then so is $A \cup A$.
- B. For any sets A and B , if $A \cup B$ is regular, then so is A .
- C. For two DFAs M_1 and M_2 , $M_1 \cup M_2$ is regular.
- ☒ D. None of the above.
- E. I don't know.

Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

Proof: Let A_1, A_2 be any two regular languages over Σ .

WTS that $A_1 \cup A_2$ is regular.

Goal: build a DFA that recognizes $A_1 \cup A_2$.

↓
accepts

Union

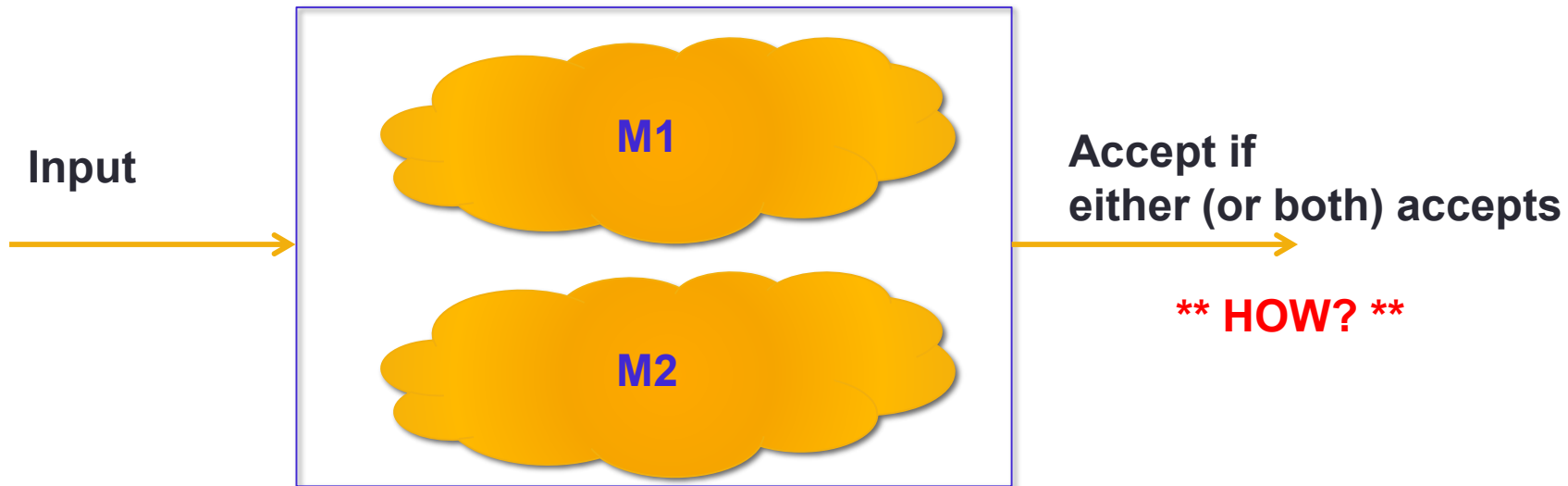
Sipser Theorem 1.25 p. 45

Goal: build a DFA that recognizes $A_1 \cup A_2$.

M_1 DFA for A_1

M_2 DFA for A_2

Strategy: use DFAs that recognize each of A_1 , A_2 .



Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

Idea: State of M is (r, y) then, r state M_1 , y state M_2 .

Proof: Let A_1, A_2 be any two regular languages over Σ .

Given $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ such that $L(M_1) = A_1$ and

$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such that $L(M_2) = A_2$ and

WTS that $A_1 \cup A_2$ is regular.

Define $M = (Q_1 \times Q_2, \Sigma, \delta, ?, ?)$. . .

Idea: run in parallel

Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

Proof: Let A_1, A_2 be any two regular languages over Σ .

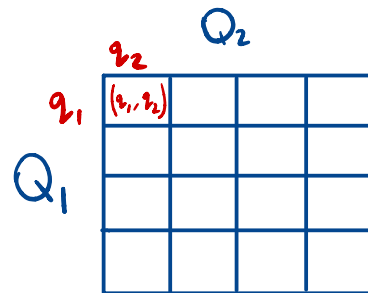
Given $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ such that $A_1 = L(M_1)$ and $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such that $A_2 = L(M_2)$

What should be the initial state of M ?

WTS that $A_1 \cup A_2$ is regular.

Define $M = (Q_1 \times Q_2, \Sigma, \delta, ?, ?)$

- A. q_0
- B. q_1
- C. q_2
- ☒ D. (q_1, q_2)
- E. I don't know.



Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

Proof: Let A_1, A_2 be any two

Given $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$:

$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such th

WTS that $A_1 \cup A_2$ is regular.

Define $M = (Q_1 \times Q_2, \Sigma, \delta, ?, ?)$

When r is a state in M_1 , s is a state in M_2 ,
and x is in Σ , then $\delta((r,s), x) =$

- A. (r,s)
- B. $(\delta(r,x), \delta(s,x))$
- C. $(\delta_1(r,x), s)$
- ☒ D. $(\delta_1(r,x), \delta_2(s,x))$
- E. I don't know.

Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

(goal: M accept w if M_1 accepts w OR M_2 accepts w)

Proof: Let A_1, A_2 be any two

Given $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$:

$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such that

WTS that $A_1 \cup A_2$ is regular.

Define $M = (\underline{Q_1 \times Q_2}, \Sigma, \delta, ?, ?)$

$$\{(q', q'') : q' \in Q_1, q'' \in Q_2\}$$

The set of accepting states for M is

- A. $F_1 \times F_2$
- B. $\{(r, s) \mid r \text{ is in } F_1 \text{ and } s \text{ is in } F_2\}$ **Intersection**
- ☒ C. $\{(r, s) \mid r \text{ is in } F_1 \text{ or } s \text{ is in } F_2\}$ **Union**
- D. $F_1 \cup F_2$
- E. I don't know.

Union

Sipser Theorem 1.25 p. 45

Proof: Let A_1, A_2 be any two regular languages over Σ .
Given $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ such that $L(M_1) = A_1$ and
 $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such that $L(M_2) = A_2$.

WTS that $A_1 \cup A_2$ is regular. Define

$M = (Q_1 \times Q_2, \Sigma, \delta, (q_1, q_2), \{(r, s) \text{ in } Q_1 \times Q_2 \mid r \text{ in } F_1 \text{ or } s \text{ in } F_2\})$
with $\delta((r, s), x) = (\delta_1(r, x), \delta_2(s, x))$ for each $(r, s) \text{ in } Q_1 \times Q_2$
and $x \text{ in } \Sigma$.

Why does $L(M) = A_1 \cup A_2$?

M simulates both machines at same time

After reading w , M is in State q .

$$(\delta^*(q_1, w), \delta^*(q_2, w))$$

State M_1 reaches on W , State M_2 reaches on W

M accepts iff :

- The first Component is accepting (So $w \in A_1$)
- or,
- The second Component is accepting (So $w \in A_2$)

So M accepts exactly the strings that are in A_1 or A_2

Therefore, $L(M) = A_1 \cup A_2$

Intersection

- *How would you prove that the class of regular languages is closed under intersection?*
- *Can you think of **more than one** proof strategy?*

$$A \cap B = \{ x \mid x \text{ in } A \text{ and } x \text{ in } B \}$$

$$A \cap B = (A^c \cup B^c)^c$$

(De Morgan's law)

A, B regular

$\Rightarrow A^c, B^c$ regular

$\Rightarrow A^c \cup B^c$ regular

$\Rightarrow (A^c \cup B^c)^c$

Payoff

$\{ w \mid w \text{ contains neither the substrings aba nor baab} \}$

Is this a regular set?

Payoff

$\{ w \mid w \text{ contains neither the substrings aba nor baab} \}$

Is this a regular set?

$A = \{ w \mid w \text{ contains aba as a substring} \}$

$B = \{ w \mid w \text{ contains baab as a substring} \}$

$$\bar{A} \cap \bar{B} = \overline{A \cup B}$$

General proof structure/strategy

Theorem: For any L over Σ , if L is regular then [the result of some operation on L] is also regular.

$$\{0\}^* = \{0^n : n \geq 0\} = \{\epsilon, 0, 00, \dots\}$$

Proof:

Given name variables for sets, DFAs assumed to exist.

WTS state goal and outline plan.

Construction using objects previously defined + new tools working towards goal. Give formal definition and explain.

Correctness prove that construction works.

Conclusion recap what you've proved.

The regular operations

Sipser Def 1.23 p. 44

For A, B languages over same alphabet, define: ① A^c

② $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$ ✓

NFAs $\left\{ \begin{array}{l} A \circ B = \{xy \mid x \in A \text{ and } y \in B\} \\ A^* = \{x_1x_2 \dots x_k \mid k \geq 0 \text{ and each } x_i \in A\} \end{array} \right.$

How can we prove that the concatenation of two regular languages is a regular language?